

Assignment #01

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS

Term: Fall 2025

Professor: Dr. Michael Caracotsios

Student Names: Pasquale Morrone, Ryan Garnett

Problem #1

Introduction

The purpose of this question is to provide an introduction to the Brayton cycle and its attributes.

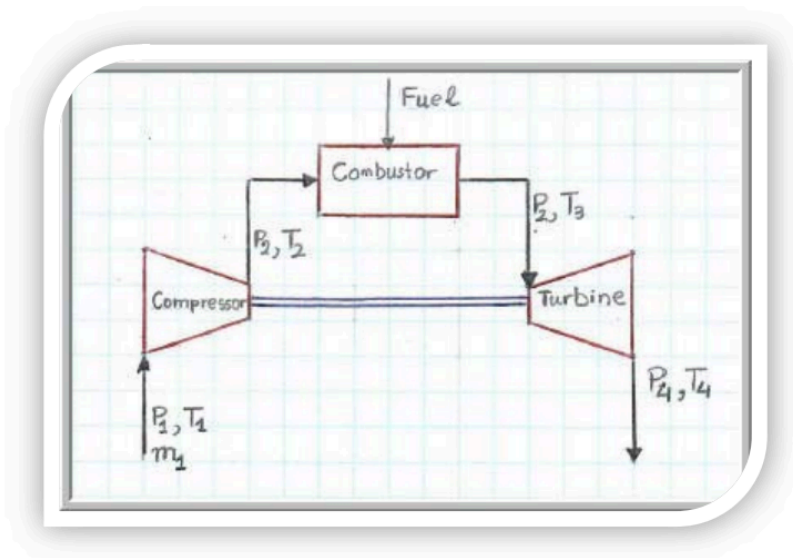
The following calculations were performed with the guidance of AI tools including Chat GPT and data for specific numerical values from google archives.

Part #	1	2	3	4	5	6	7	8
Solution	16.943	531 MMSCFD	55.40%	6.86	55.4%	$\dot{W} = \dot{m} \text{ (Kg/S) } * 0.4243 \text{ mw}$	80.10%	33321.28 lb/hr of kerosene

Known Values:

- **P1 / P4:** 100 Kpa
- **T1:** 303.15 K
- **T3:** 1523.15 K

Figure A: Power Production: Brayton Cycle



Calculations

Part 1: The pressure ratio P_2 / P_1 that maximizes the net power generation

$$r_p = \left(\frac{T_3}{T_1} \right)^{\frac{k}{2(k-1)}}, \quad k = \frac{c_p}{c_v}$$

$$k = \frac{1.005 \text{ kJ/kg} \cdot \text{K}}{0.718 \text{ kJ/kg} \cdot \text{K}} = 1.399 \text{ kJ/kg} \cdot \text{K}$$

$$r_p = \left(\frac{1523.15 \text{ K}}{303.15 \text{ K}} \right)^{\left(\frac{1.399 \text{ kJ/kg} \cdot \text{K}}{2 \times (1.399 \text{ kJ/kg} \cdot \text{K} - 1)} \right)} = 16.943$$

Part 2: The minimum air flow rate in MMSCFD required to produce 100 MW of power

solve for T2 & T4

$$T_2 = T_1 \times r_p^{\left(\frac{k-1}{k}\right)} \quad , \quad T_4 = T_3 \times r_p^{-\left(\frac{k-1}{k}\right)}$$

$$T_2 = 303.15K \times 16.943^{\left(\frac{1.399 \text{ kJ/kg} \cdot K - 1}{1.399 \text{ kJ/kg} \cdot K}\right)} = 679.36K$$

$$T_4 = 1523.15K \times 16.943^{-\left(\frac{1.399 \text{ kJ/kg} \cdot K - 1}{1.399 \text{ kJ/kg} \cdot K}\right)} = 679.56K$$

Solve for specific work values

$$w_c = c_p \times (T_2 - T_1) \quad , \quad w_t = c_p \times (T_3 - T_4)$$

$$w_c = 1.005 \text{ kJ/kg} \cdot K \times (679.36 K - 303.15 K) = 378.09 \text{ kJ/kg}$$

$$w_t = 1.005 \text{ kJ/kg} \cdot K \times (1523.15 K - 679.56 K) = 847.807 \text{ kJ/kg}$$

Required mass flow to reach 100 mw

$$\dot{m} = \frac{\text{power}}{w_{net}} \quad , \quad w_{net} = w_t - w_c$$

$$w_{net} = 847.807 \text{ kJ/kg} - 378.09 \text{ kJ/kg} = 469.717 \text{ kJ/kg}$$

$$\dot{m} = \frac{100 \times 10^6 \text{ W}}{469,717 \text{ J/kg}} = 212.89 \text{ kg/s}$$

Convert m to standard volumetric flow

$$v_{std} = \frac{RT_{std}}{P_{std}} = \frac{287.058 \text{ J/kg} \cdot K \times 288.706 K}{101325 \text{ Pa}} = 0.8175 \text{ m}^3/\text{kg}$$

$$\dot{v}_{std} = 212.89 \text{ kg/s} \times 0.8175 \text{ m}^3/\text{kg} = 174.04 \text{ m}^3/\text{s}$$

Conversion to MMSCFD

$$174.04 \text{ m}^3/\text{s} \times 35.3146667 \text{ ft}^3/\text{m}^3 \times \frac{86400}{10^6} = 531 \text{ MMSCFD}$$

Part 3: The cycle efficiency at maximum power

Cycle efficiency at maximum power

$$\eta_{max\ power} = 1 - \sqrt{\frac{T_1}{T_3}}$$

$$\eta_{max\ power} = 1 - \sqrt{\frac{303.15\ K}{1523.15\ K}} = 0.5539 \approx 55.4\%$$

Part 4: The pressure ratio P /P 2 1 that maximizes the cycle

efficiencyThe pressure ratio P /P 2 1 that maximizes the cycle efficiency

*AI was used to add a regenerator to address the problem because to my knowledge you cannot solve for the desired value with ideal Brayton without one.

$$\eta(r_p) = \frac{T_3 - T_3 r_p^{-a} - T_1 r_p^a + T_1}{T_3 - T_1 r_p^a - \varepsilon(T_3 r_p^{-a} - T_1 r_p^a)}$$

rp (dimensionless)	T2 = T1*rp^a (K)	T4 = T3*rp^(-a) (K)	Numerator = (T3-T4)-(T2-T1) (K)	Denominator = (T3-T2)-ε*(T4-T2) (K)	Efficiency η (dimensionless)
6	505.8089	912.8801	407.611	691.684	0.589302
6.5	517.5097	892.2401	416.55	705.856	0.590135
6.865139415	525.6543	878.4155	422.2301	715.2867	0.590295
7	528.5842	873.5466	424.1692	718.5959	0.590275
8	549.1403	840.847	436.3128	740.6444	0.589099

Optimized pressure ratio (when ε = 0.8)

$$r_{p,opt} = 6.865$$

Part 5: The value of the maximum efficiency

$$\eta_{(rp,\varepsilon)} = \frac{(T_3 - T_4) - (T_2 - T_1)}{(T_3 - T_2) - \varepsilon(T_4 - T_2)}$$
$$\eta_{(rp,\varepsilon)} = \frac{(1,523.15 \text{ K} - 679.56 \text{ K}) - (679.36 \text{ K} - 303.15 \text{ K})}{(1,523.15 \text{ K} - 679.36 \text{ K}) - 0.8(679.56 \text{ K} - 679.36 \text{ K})} = 0.554$$
$$\eta_{(rp,\varepsilon)} = 0.554 \approx 55.4\%$$

Part 6: The net power in MW generated at maximum efficiency

$$a = \frac{k - 1}{k} = \frac{1.4 - 1}{1.4} = 0.2857$$

$$T_2 = T_1 r_p^a = 303.15 \text{ K} \times 6.86^{0.2857} = 525.52 \text{ K}$$

$$T_4 = T_3 r_p^{-a} = 1,523.15 \text{ K} \times 6.86^{-0.2857} = 878.627 \text{ K}$$

$$w_c = c_p (T_2 - T_1) = 1,005 \text{ J/kg} \cdot \text{K} \times (525.52 \text{ K} - 303.15 \text{ K}) = 223,481.85 \text{ J/kg}$$

$$w_t = c_p (T_3 - T_4) = 1,005 \text{ J/kg} \cdot \text{K} \times (1,523.15 \text{ K} - 878.627 \text{ K}) = 647,745.615 \text{ J/kg}$$

$$w_{net} = 424.265 \text{ kJ/kg}$$

$$\dot{w} = \dot{m} \times w_{net}$$

$$\dot{w} = \dot{m} (\text{kg/s}) \times 0.4243 \text{ mw}$$

Maximum $\dot{m}$ is set by constraints based on the physical hardware the system is constructed from so the closest I can get to the requested data at this time is the formula above.

Part 7: The Carnot efficiency of the cycle

$$\eta_{carnot} = 1 - \frac{T_c}{T_h}$$

$$T_h = T_3 = 1,523.15 \text{ K}$$

$$T_c = T_1 = 303.15 \text{ K}$$

$$\eta_{carnot} = 1 - \frac{303.15 \text{ K}}{1,523.15 \text{ K}} = 0.801$$

$$\eta_{carnot} = 80.1\%$$

Part 8: If the fuel used in the combustor is kerosene what is the amount in lbs/hr required for maximum power production

$$\dot{Q}_{in} = \dot{m}_{fuel} \times LHV, \quad \dot{m}_{fuel} = \frac{\dot{Q}_{in}}{LHV}$$

$$q_{in} = c_p \times (T_3 - T_2) = 1.005 \text{ KJ/kg} \times (1,523.15 \text{ K} - 679.36 \text{ K}) = 848 \text{ KJ/kg air}$$

$$\dot{Q}_{in} = \dot{m}_{air} \times q_{in} = 212.89 \text{ kg/s} \times 848 \text{ kJ/kg air} = 180,532.62 \text{ kJ/s}$$

$$\text{For kerosene, } LHV = 43,000 \text{ kJ/kg}$$

$$\dot{m}_{fuel} = \frac{180,532.62 \text{ kJ/s}}{43,000 \text{ kJ/kg}} = 4.19 \text{ kg/s}$$

$$\dot{m}_{fuel} = 4.19 \text{ kg/s} \times 7,936.6 = 33,321.28 \text{ lb/hr of kerosene}$$

Problem #2 - Vapor Fraction Methods

Introduction

The goal of this problem is to solve for the vapor fraction (α) in a multicomponent mixture. Numerical methods (Newton and Secant) were applied to find the root of the nonlinear equation.

Solution:

Given Equation:
$$F(\alpha) = \sum_{i=1}^{14} \frac{z_i(K_i - 1)}{1 + \alpha(K_i - 1)} = 0$$

Knowns:

- z_i = the overall mole fraction of the component i
- K_i = equilibrium constant
- α = vapor fraction

Newton Method Iteration

First we need to derive the given equation in a way where it is suitable to use the Newton Method, this was done by ChatGPT as a tool to quickly derive a larger function such as this.

$$\frac{dF}{d\alpha} = - \sum_{i=1}^{14} \frac{z_i(K_i - 1)^2}{[1 + \alpha(K_i - 1)]^2}$$

Then we must update the formula and complete the iterations.

Updated Formula:
$$\alpha_{n+1} = \alpha_n - \frac{F(\alpha_n)}{F'(\alpha_n)}$$

Starting Guess: $\alpha_0 = 0.500$

Noticing that our z values do not add up to 1, I thought it best to normalize them so then we can get mole fractions. This was done by directly asking ChatGPT for the formula to do so.

In order to normalize we must take the z values given and divide them by the sum of all the z values.

$$z_{i, \text{ normalized}} = \frac{z_i}{\sum_{i=1}^{14} z_i}$$

1																			
2	alpha (a)		0.5	Guess															
3																			
4	i	z(i)	z(i) normal	K(i)	K(i) - 1	1 + a*(K - 1z*(K-1)/de	derm: -z*(K-1)^2/den^2												
5	1	0.1176	0.14756	49.9058	48.9058	25.4529	0.28353	-0.54478											
6	2	0.00278	0.00348	13.4458	12.4458	7.22288	0.006	-0.01034											
7	3	0.00344	0.00431	3.47491	2.47491	2.23746	0.00477	-0.00527											
8	4	0.00317	0.00398	1.28255	0.28255	1.14127	0.00099	-0.00024											
9	5	0.00119	0.00149	0.6282	-0.3718	0.8141	-0.00068	-0.00031											
10	6	0.00079	0.001	0.47768	-0.52232	0.73884	-0.0007	-0.0005											
11	7	0.03397	0.04263	0.06442	-0.93558	0.53221	-0.07493	-0.13173											
12	8	0.04467	0.05605	0.05558	-0.94442	0.52779	-0.10029	-0.17946											
13	9	0.00257	0.00323	0.22687	-0.77314	0.61343	-0.00407	-0.00513											
14	10	0.06565	0.08238	0.18357	-0.81643	0.59179	-0.11365	-0.15679											
15	11	0.41228	0.51732	0.09271	-0.90729	0.54635	-0.85909	-1.42663											
16	12	0.03258	0.04088	0.13672	-0.86329	0.56836	-0.0621	-0.09432											
17	13	0.02256	0.02831	0.07225	-0.92775	0.53613	-0.04899	-0.08478											
18	14	0.0537	0.06738	0.03823	-0.96177	0.51912	-0.12484	-0.23129											
19	sum	0.79694			F(a)	-1.09406													
20					F'(a)	-2.87156													

Then we simply prepare a table for the Newton method to be completed where we use a lot of the built in excel function SUMPRODUCT to calculate alpha values and its values in respect to the function F(α) at different n (iterations). This was repeated until we reached the desired product.

Excel Results:

- $\alpha_0 = 0.5000$, $F(\alpha_0) = -1.09406$
- $\alpha_1 = 0.119$, $F(\alpha_1) = 0.23386$
- $\alpha_2 = 0.14634$, $F(\alpha_2) = 0.03362$
- $\alpha_3 = 0.15167$, $F(\alpha_3) = 0.00086$
- $\alpha_4 = 0.15181$, $F(\alpha_4) \approx 0$

22																			
23	Newton	n	a(n)	F(a,n)	F'(a,n)	delta a	a (n+1)												
24		0	0.5	-1.09406	-2.87156	-0.381	0.119												
25		1	0.119	0.23386	-8.55606	0.02733	0.14634												
26		2	0.14634	0.03362	-6.30152	0.00534	0.15167												
27		3	0.15167	0.00086	-5.98466	0.00014	0.15181												
28		4	0.15181	5.8E-07	-5.97655	9.7E-08	0.15181												
29		5	0.15181	2.7E-13	-5.97655	4.5E-14	0.15181												
30		6	0.15181	1.7E-16	-5.97655	2.8E-17	0.15181												
31		7	0.15181	-5.6E-17	-5.97655	-9.3E-18	0.15181												
32		8	0.15181	-5.6E-17	-5.97655	-9.3E-18	0.15181												
33		9	0.15181	-5.6E-17	-5.97655	-9.3E-18	0.15181												
34		10	0.15181	-5.6E-17	-5.97655	-9.3E-18	0.15181												

Final Newton Result: $\alpha = 0.15181$

Secant Method Iteration

We once again are using the same given formula for the secant method, and we are also using the same formula provided by ChatGPT to normalize our z numbers.

First we will choose two starting guesses which will act as α_0 and α_1 respectively on the interval (0,1).

Guesses: $\alpha_0 = 0$ $\alpha_1 = 0.5$

Update the given Equation for the secant method (this was calculated by ChatGPT)

$$\alpha_{k+1} = \alpha_k - F(\alpha_k) \frac{\alpha_k - \alpha_{k-1}}{F(\alpha_k) - F(\alpha_{k-1})}$$

We stop iterating the function once we get to a number that is very close to the goal, which in our case is 0.

Excel Results:

Secant Method					
k	a(k-1)	a(k)	F(a(k-1))	F(a(k))	a(k+1)
1	0	0.5	6.5124	-1.09406	0.42808
2	0.5	0.42808	-1.09406	-0.89896	0.09672
3	0.42808	0.09672	-0.89896	0.45648	0.20832
4	0.09672	0.20832	0.45648	-0.27081	0.16676
5	0.20832	0.16676	-0.27081	-0.08357	0.14822
6	0.16676	0.14822	-0.08357	0.02186	0.15206
7	0.14822	0.15206	0.02186	-0.00149	0.15182
8	0.15206	0.15182	-0.00149	-2.5E-05	0.15181
9	0.15182	0.15181	-2.5E-05	3E-08	0.15181
10	0.15181	0.15181	3E-08	-6E-13	0.15181

So as you can see we got the same resulting alpha value for both the secant and newton methods. That value being $\alpha = 0.15181$ or **15.181 % Vapor by mole.**

Checking Result Using GoalSeek:

To do this we pick one cell and write in a value that makes sense as a guess for alpha (I put 0.5). Then in the cell next to it we enter the general formula that we have been referencing for the majority of this problem, but for our alpha value we use the guess we just made for the reference cell. Then we go through excel's GoalSeek process where we choose the cell where the function is written to be the "Set Cell", we set the value to 0, and finally we complete the calculation by changing the cell we made the guess in.

GoalSeek	
alpha	F(a)
0.15181	5.3E-05

Conclusion:

The final vapor fraction came out to be about $\alpha = 0.15181$, which means the mixture is roughly 15% vapor and 85% liquid by mole. Results for the Secant method and Newton's method agreed on this value, and Goal Seek confirmed it. The main assumption is that the mixture behaves ideally since it was not explicitly stated for this problem. A key source of error could be not normalizing the z_i values, which would shift the answer. But overall the methods lined up with one another, and the result seems reasonable.